

Lecture 3

The exterior derivative, Stokes, and the symplectic form

Objectives. Course map: G1, step 3 of 9. *Calculus on forms, and the one 2-form mechanics is built on. The exterior derivative d tells a closed form from an exact one. Stokes' theorem, cited, then shows that the integral of a closed 1-form round a loop depends only on the loop's class, which is what will make the action variables of [Lecture 6](#) well defined. The second half names the symplectic form, a closed, non-degenerate 2-form, and checks that its sign convention gives Hamilton's equations.*

3.1 The question

[Lecture 2](#) gave $p \cdot dq$ a type, a 1-form, and left two requirements on the loop integral (2.8) open: it must not change when the loop is reparametrised, and it must not change when the loop is deformed. So the question of this lecture is

when does the integral of a 1-form round a loop depend only on the loop's class — and what structure on phase space does $p \cdot dq$ come from?

The answer to the first is *closedness*, with Stokes' theorem ([Corollary 3.13](#)). The answer to the second is the symplectic form ω , of which $p \cdot dq$ is a primitive up to sign (§ 3.6).

3.2 The exterior derivative

We have d on functions. Extending it to forms of every degree is what turns this algebra into a calculus.

Definition 3.1 (Exterior derivative). On a 0-form (a function), df is [Definition 2.8](#). On a 1-form $\alpha = \sum_i a_i dx^i$,

$$d\alpha = \sum_i da_i \wedge dx^i = \sum_{i,j} \frac{\partial a_i}{\partial x^j} dx^j \wedge dx^i, \quad (3.1)$$

and in general d acts on $\sum a_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}$ by differentiating each coefficient and wedging d of it on the left. It raises degree by one and obeys the graded Leibniz rule

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta, \quad \alpha \text{ of degree } k. \quad (3.2)$$

The formula is written in coordinates, so it is fair to ask whether a different chart would give a different answer. It would not: d is characterised by (3.2), by agreeing with

Definition 2.8 in degree zero, and by $d \circ d = 0$, and those three conditions mention no coordinates. We take the characterisation on trust (*cited*; da Silva, §6) and use the formula.

In the plane, for $\alpha = P dx + Q dy$, (3.1) gives

$$d\alpha = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \wedge dy, \quad (3.3)$$

using $dx \wedge dx = 0$ and $dy \wedge dx = -dx \wedge dy$. Evaluated on a small parallelogram, $d\alpha$ is the circulation of α around its boundary (Figure 3.1). That reading is Stokes' theorem in miniature, and § 3.4 does no more than add such squares up.

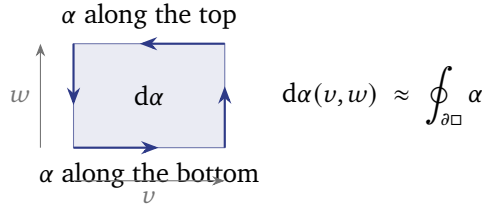


Figure 3.1: $d\alpha$ on an infinitesimal parallelogram is the circulation of α around its boundary. The four edge contributions do not cancel exactly, and what is left over is $d\alpha$.

Theorem 3.2 ($d \circ d = 0$). $d(d\alpha) = 0$ for every form α .

Proof. It is enough to check it on a function, since the general case follows by (3.2) and $d(dx^i) = 0$. With $\alpha = f$,

$$d(df) = d\left(\sum_i \frac{\partial f}{\partial x^i} dx^i\right) = \sum_{i,j} \frac{\partial^2 f}{\partial x^j \partial x^i} dx^j \wedge dx^i.$$

Now split the double sum into the pairs $\{i, j\}$. Within each pair the coefficient $\partial^2 f / \partial x^j \partial x^i$ is *symmetric* in i, j because mixed partials of a smooth function commute, while $dx^j \wedge dx^i$ is *antisymmetric* by (2.13). Within each pair the two terms are equal and opposite, so the sum vanishes pair by pair. ■

Definition 3.3 (Closed, exact). A k -form α is *closed* if $d\alpha = 0$, and *exact* if $\alpha = d\beta$ for some $(k-1)$ -form β .

Theorem 3.2 says exact $\Rightarrow$ closed. The converse is the interesting direction, and it is *local*.

Theorem 3.4 (Poincaré lemma · *cited*). A closed form on a ball — or on any region that can be smoothly contracted to a point — is exact. (da Silva, §6; Arnold, §36.)

Everything this course does with forms lives in the gap between Theorem 3.2 and Theorem 3.4: closed and exact differ, and they differ only because of the shape of the region (Figure 3.2). § 3.3 is the smallest example, and Lecture 6 is the payoff.

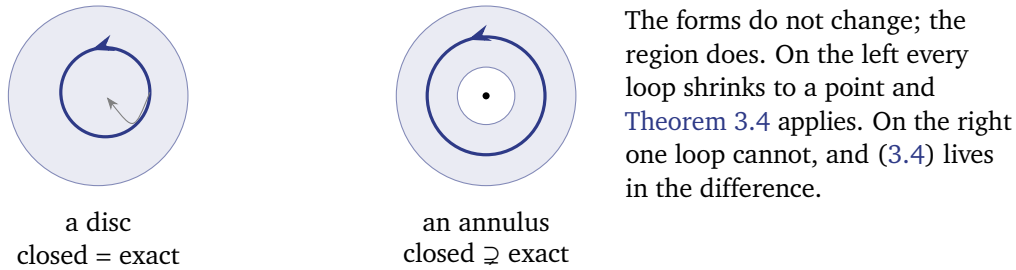


Figure 3.2: Closed versus exact is a statement about the *region*. Locally the two notions agree; globally they are separated by exactly the loops that fail to bound.

Reading

Grad, curl and div are one operator

In $\mathbb{R}^3$ the forms of each degree have three, three and one independent components — and so do the classical vector operators. Writing d in each degree reproduces them:

degree	object	d gives	classical name
0	f	$\sum_i \partial_i f \, dx^i$	grad
1	$Pdx + Qdy + Rdz$	a 2-form with components $\partial_y R - \partial_z Q, \dots$	curl
2	$Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy$	$(\partial_x P + \partial_y Q + \partial_z R) \, dx \wedge dy \wedge dz$	div

Read in this table, [Theorem 3.2](#) is two familiar identities at once: $\text{curl grad} = 0$ and $\text{div curl} = 0$. They were never two facts. The dictionary is special to three dimensions — it relies on 3-forms and 1-forms both having three components — which is the reason the classical operators do not generalise and d does. Phase space is $2n$ -dimensional, so from here on only d is available — which is why this comparison is worth making once and then leaving behind.

3.3 Closed but not exact: the annulus

Example 3.5 (The angle form). On the punctured plane $\mathcal{M} = \mathbb{R}^2 \setminus \{0\}$ — an annulus, for present purposes — put

$$\alpha = \frac{x \, dy - y \, dx}{x^2 + y^2}. \quad (3.4)$$

Away from the origin this is smooth, and a two-line computation with (3.3) gives $d\alpha = 0$: it is *closed*.

In polar coordinates $\alpha = d\theta$, which makes it look exact. It is not, and the reason is that θ is not a function on the annulus: continuing it once around the hole returns it increased by 2π . The name $d\theta$ is a name for the 1-form (3.4), not an assertion that something has been differentiated. To see this is not a quibble, integrate.

Proposition 3.6. α is closed and not exact on $\mathbb{R}^2 \setminus \{0\}$.

Proof. Closedness is [Example 3.5](#). For the rest, let γ be the unit circle traversed once anticlockwise, $\gamma(t) = (\cos t, \sin t)$, $t \in [0, 2\pi]$. Substituting into (3.4) gives $\oint_{\gamma} \alpha = \int_0^{2\pi} dt = 2\pi$. If α were df for a function f on the annulus, then by [Proposition 3.10](#) below the integral around any closed curve would be $f(\gamma(2\pi)) - f(\gamma(0)) = 0$. It is 2π , so no such f exists. ■

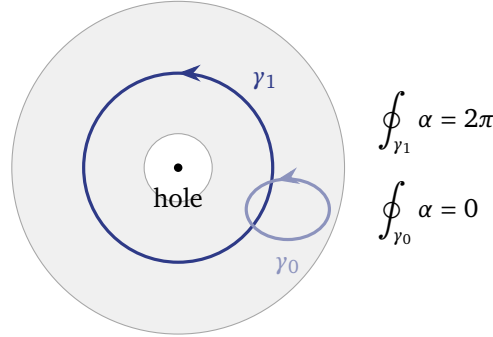


Figure 3.3: Two loops in the annulus. γ_0 can be shrunk to a point inside the region and integrates to zero; γ_1 cannot, because the hole is in the way, and integrates to 2π . The obstruction to exactness is the hole, and nothing else.

This is the smallest interesting fact in the lecture, and the one to remember. It is tempting to read [Proposition 3.6](#) as bookkeeping. It is not:

on the invariant tori of [Lecture 5](#), in $\mathbb{R}^{2n}$ or a cotangent bundle, $p \cdot dq$ is closed, which is what makes (2.8) depend on the loop only through its class; and it need not be exact there, which is what allows the answer to be nonzero.

The two halves do different jobs and neither can be dropped. Were $p \cdot dq$ exact on the torus, (2.8) would give $I = 0$ for every cycle on it, by [Proposition 3.10](#). The nonzero actions of the oscillator, the pendulum and the Kepler problem exist for the reason [Figure 3.3](#) exhibits — a cycle that does not bound, and a closed form with no primitive along it.

3.4 Pullback, integration, and Stokes' theorem

Pullback

Forms have one property vectors do not: they can be transported *backwards* along any smooth map, with no assumption that the map be invertible.

Definition 3.7 (Pullback). Let $F: N \rightarrow \mathcal{M}$ be smooth and let α be a 1-form on $\mathcal{M}$. Its *pullback* $F^*\alpha$ is the 1-form on N given by

$$(F^*\alpha)_p(v) = \alpha_{F(p)}(dF_p v), \quad (3.5)$$

where $dF_p: T_p N \rightarrow T_{F(p)} \mathcal{M}$ is the derivative of F . For a 2-form, $(F^*\beta)_p(v, w) = \beta_{F(p)}(dF_p v, dF_p w)$, and similarly in any degree.

In practice pullback is substitution: if F is given by $x^i = x^i(u)$, then $F^*(dx^i) = \sum_a (\partial x^i / \partial u^a) du^a$, and one simply substitutes. Three rules follow, used later and the reason the formalism is worth its cost:

$$F^*(df) = d(f \circ F), \quad F^*(\alpha \wedge \beta) = F^*\alpha \wedge F^*\beta, \quad F^*(d\alpha) = d(F^*\alpha). \quad (3.6)$$

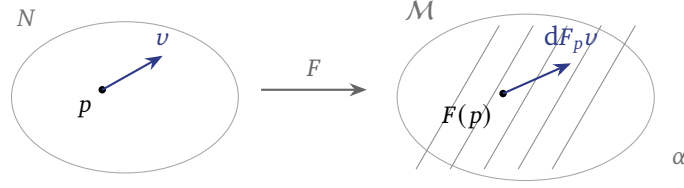


Figure 3.4: Pullback. The vector is pushed forward by dF , the form is evaluated where it lives, and the answer is recorded back at p . Nothing needs to be inverted, which is why F^* exists for every smooth F .

The first is the chain rule, and the second is the definition applied pointwise. The third follows from them for a 1-form, the case the deformation argument below uses: with $\alpha = \sum_i a_i dx^i$ and $F^*\alpha = \sum_i (a_i \circ F) d(x^i \circ F)$, the graded Leibniz rule (3.2) and $dd(x^i \circ F) = 0$ give $d(F^*\alpha) = \sum_i d(a_i \circ F) \wedge d(x^i \circ F) = F^*(\sum_i da_i \wedge dx^i) = F^*(d\alpha)$. The last rule — d commutes with pullback — is what makes “closed” and “exact” coordinate-independent notions, and [Lecture 4](#) will use the middle one to recognise canonical transformations as the maps satisfying $F^*\omega = \omega$.

Integration over a curve

Now the type-checking of § 2.4 pays off.

Definition 3.8 (Line integral of a 1-form). For a smooth curve $\gamma: [a, b] \rightarrow \mathcal{M}$ and a 1-form α ,

$$\int_{\gamma} \alpha = \int_a^b \gamma^* \alpha = \int_a^b \alpha_{\gamma(t)}(\dot{\gamma}(t)) dt. \quad (3.7)$$

Proposition 3.9 (Reparametrisation invariance). If $\varphi: [c, d] \rightarrow [a, b]$ is smooth, increasing and onto, then $\int_{\gamma \circ \varphi} \alpha = \int_{\gamma} \alpha$.

Proof (★). By the chain rule $\frac{d}{ds}(\gamma \circ \varphi) = \dot{\gamma}(\varphi(s)) \varphi'(s)$, so the integrand acquires exactly the factor $\varphi'(s)$ that the substitution rule removes. ■

So the answer depends on the curve and its direction, not on the schedule. That was requirement 2 of § 2.4, and [Definition 3.8](#) met it by construction: a 1-form is precisely the kind of object whose integral over a curve needs no further data.

Proposition 3.10 (Fundamental theorem). $\int_{\gamma} df = f(\gamma(b)) - f(\gamma(a))$. In particular $\oint_{\gamma} df = 0$ for every closed curve.

Proof. $df(\dot{\gamma}) = \frac{d}{dt}(f \circ \gamma)$ by (2.9), and the one-variable fundamental theorem of calculus does the rest. ■

[Proposition 3.10](#) is what [Proposition 3.6](#) used, and it is also the statement that an exact form carries no information about loops. Everything interesting therefore lies with forms that are closed and not exact.

Stokes' theorem

Theorem 3.11 (Stokes · cited). *Let S be a compact oriented surface with boundary ∂S , carrying the induced orientation, and let α be a 1-form defined on a neighbourhood of S . Then*

$$\int_S d\alpha = \oint_{\partial S} \alpha. \quad (3.8)$$

The same identity holds in every degree, for a k -form on a $(k+1)$ -dimensional region. (Proof: da Silva, §6; Arnold, §36.)

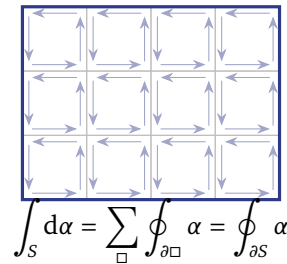
We do not prove it. What we do instead is say exactly why it is true, because the mechanism is short and is the picture the rest of Part I uses.

Reading

Example 3.12 (Stokes on a rectangle, verified). Let $S = [x_0, x_1] \times [y_0, y_1]$ and $\alpha = P dx + Q dy$. By (3.3) and Fubini,

$$\int_S d\alpha = \int_{y_0}^{y_1} \int_{x_0}^{x_1} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \int_{y_0}^{y_1} [Q]_{x_0}^{x_1} dy - \int_{x_0}^{x_1} [P]_{y_0}^{y_1} dx,$$

after doing one integral in each term by the fundamental theorem of calculus. The four resulting boundary terms are exactly the four edges of $\oint_{\partial S} \alpha$, each with the sign the anticlockwise orientation gives it. That is (3.8) for a rectangle.



every interior edge is
traversed twice, oppositely
only ∂S survives

Figure 3.5: Why Stokes is true. Chop S into cells; on each, $d\alpha$ is the circulation around its boundary (Figure 3.1). Summing, every interior edge is traversed once in each direction and cancels, leaving the outer boundary alone.

The payoff: integrals that depend only on the class of the loop

Corollary 3.13 (Deformation invariance). *Let α be a closed 1-form on a region U , and let $F: S^1 \times [0, 1] \rightarrow U$ be a smooth deformation — a family of closed curves $\gamma_s = F(\cdot, s)$ inside U , running from γ_0 to γ_1 . Then*

$$\oint_{\gamma_1} \alpha = \oint_{\gamma_0} \alpha.$$

Proof. Pull α back to the cylinder $C = S^1 \times [0, 1]$, where it is a 1-form on a 2-dimensional region. It is still closed, because d commutes with pullback (3.6): $d(F^*\alpha) = F^*(d\alpha) = 0$. The boundary of C is its two ends, the top with the orientation induced from C and the bottom with the opposite one, so $\partial C = S^1 \times \{1\} - S^1 \times \{0\}$. By Theorem 3.11 applied on C ,

$$\oint_{\gamma_1} \alpha - \oint_{\gamma_0} \alpha = \oint_{\partial C} F^*\alpha = \int_C d(F^*\alpha) = 0. \quad \blacksquare$$

This is requirement 3 of § 2.4, discharged for closed forms: the integral of a closed 1-form depends on the loop only through its class. That $p \cdot dq$ is closed on the invariant tori, in $\mathbb{R}^{2n}$ or a cotangent bundle, is [Lecture 5](#)'s. An action variable is attached not to a curve but to a *class* of curves, and [Lecture 5](#) will produce exactly n such classes on an n -dimensional invariant torus — the basis of cycles $\gamma_1, \dots, \gamma_n$ that [Figure 3.6](#) shows for $n = 2$, and [Lecture 6](#) will integrate $p \cdot dq$ over each of them in turn.

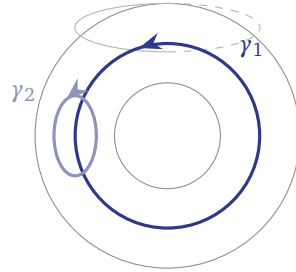


Figure 3.6: The two independent cycles on T^2 , drawn on the torus seen from above. Neither bounds, so a closed 1-form may integrate to something nonzero over each, and [Lecture 6](#) calls the two answers $2\pi I_1$ and $2\pi I_2$. This picture is the target of the first half of the lecture; [Lecture 5](#) supplies the torus.

The worked example: the oscillator's loop integral

Take the oscillator at unit frequency, $H = \frac{1}{2}(p^2 + q^2)$, and its orbit $H = E > 0$: the circle of radius $\sqrt{2E}$, traversed clockwise ([§ 1.4](#)), $q = \sqrt{2E} \cos t$, $p = -\sqrt{2E} \sin t$ for $t \in [0, 2\pi]$. Then

$$\oint_{\gamma} p \, dq = \int_0^{2\pi} (-\sqrt{2E} \sin t)(-\sqrt{2E} \sin t) \, dt = 2\pi E. \quad (3.9)$$

Stokes gives the same number without a parametrisation. The clockwise orbit bounds the disc D it encloses, but with the orientation opposite to the one $dq \wedge dp$ gives D ; and $d(p \, dq) = dp \wedge dq = -dq \wedge dp$. The two signs cancel:

$$\oint_{\gamma} p \, dq = - \int_D d(p \, dq) = \int_D dq \wedge dp = \text{area of } D = 2\pi E.$$

The loop integral is the area enclosed, and it is positive. [Lecture 6](#) calls $\frac{1}{2\pi}$ of it the oscillator's action, $I = E$.

3.5 The symplectic form

[Lecture 2](#) built 2-forms, and [§ 3.2](#) showed how to tell a closed one. The phase space of mechanics carries one closed 2-form with a special property, and the rest of Part I is built on it. One piece of notation first: a 2-form eats two vectors, and feeding it one leaves a 1-form.

Definition 3.14 (Interior product). For a 2-form β and a vector v at the same point, $\iota_v \beta$ is the 1-form $w \mapsto \beta(v, w)$. The slot that v fills is the *first*.

Definition 3.15 (Symplectic form, symplectic manifold). A 2-form ω on a manifold $\mathcal{M}$ is *symplectic* if

1. it is *closed*: $d\omega = 0$; and
2. it is *non-degenerate*: at each point, if $\omega(v, w) = 0$ for every w , then $v = 0$.

The pair (M, ω) is a *symplectic manifold*.

Condition 2 says that the linear map $v \mapsto \iota_v \omega$ from $T_p M$ to $T_p^* M$ has zero kernel. The two spaces have the same dimension, so

$$v \mapsto \iota_v \omega \quad \text{is an isomorphism } T_p M \xrightarrow{\sim} T_p^* M. \quad (3.10)$$

A symplectic form converts vectors into covectors and back, with no choices made — the job an inner product does in Euclidean space, done by an antisymmetric object.

Example 3.16 (The standard symplectic form). On $\mathbb{R}^{2n}$ with coordinates $q_1, \dots, q_n, p_1, \dots, p_n$,

$$\omega = \sum_{i=1}^n dq_i \wedge dp_i, \quad \omega(v, w) = \sum_{i=1}^n (v^{q_i} w^{p_i} - v^{p_i} w^{q_i}), \quad (3.11)$$

by (2.12). Geometrically, $\omega(v, w)$ is the sum over i of the signed areas of the shadows of the parallelogram on v, w in the (q_i, p_i) -planes; for $n = 1$ it is the signed area itself (Figure 3.7). It pairs each position only with its own momentum.

In matrix form, with vectors written as columns in the order (q, p) ,

$$\omega(v, w) = v^T \Omega w, \quad \Omega = \begin{pmatrix} 0 & \mathbb{1}_n \\ -\mathbb{1}_n & 0 \end{pmatrix}. \quad (3.12)$$

It is closed because its coefficients are constant, and non-degenerate because $\det \Omega = 1$.

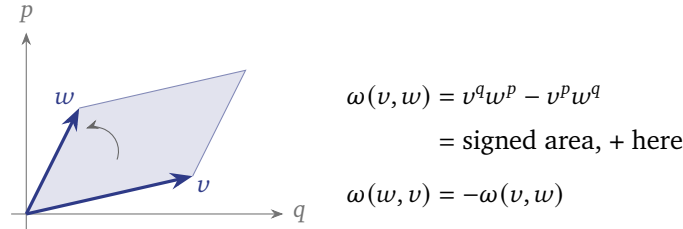


Figure 3.7: On one degree of freedom ω is the signed area in the (q, p) -plane — Figure 2.5 with q, p for x, y . For n degrees of freedom it adds n such areas, one in each plane (q_i, p_i) .

The sign test, on one degree of freedom

The wedge order $dq \wedge dp$ is not a matter of taste. Ask which X satisfies $\iota_X \omega = dH$ on $\mathbb{R}^2$. By Definition 3.14 and (2.12),

$$\iota_X (dq \wedge dp) = dq(X) dp - dp(X) dq = X^q dp - X^p dq,$$

and matching with $dH = H_q dq + H_p dp$ gives

$$X^q = \frac{\partial H}{\partial p}, \quad X^p = -\frac{\partial H}{\partial q} : \quad (3.13)$$

Hamilton's equations, both signs right; for the pendulum $H = \frac{1}{2}p^2 - \cos q$ this is $\dot{q} = p$, $\dot{p} = -\sin q$. Had we written $dp \wedge dq$, the same computation would return $X^q = -H_p$ and $X^p = +H_q$ — both signs wrong. That is the whole reason for the order.

3.6 What it buys

- *The action variables can be defined.* A closed 1-form integrates to the same number over any two loops that deform into one another (Corollary 3.13). On the invariant tori of Lecture 5, in $\mathbb{R}^{2n}$ or a cotangent bundle, $p \cdot dq$ is closed, and Lecture 6 integrates it over each torus's n independent cycles (Figure 3.6).
- $p \cdot dq$ and ω are one object. On $T^*\mathbb{R}^n$, by (3.1) and (2.13),

$$d\left(\sum_i p_i dq_i\right) = \sum_i dp_i \wedge dq_i = -\omega,$$

so the standard symplectic form is exact, and closed by Theorem 3.2. The minus sign comes with the wedge order $dq \wedge dp$ that the sign test fixed. Lecture 4 shows the same on every cotangent bundle, where $p \cdot dq$ needs no coordinates.

- *A symplectic form turns functions into flows.* By (3.10), $\iota_X \omega = dH$ has exactly one solution X , and on $\mathbb{R}^2$ it is Hamilton's vector field (3.13). Lecture 4 makes this the definition of X_H on any symplectic manifold, and shows that it agrees with Lecture 2's.

3.7 Caveats, in one place

- *Status.* Three results are cited: that d does not depend on the chart, the Poincaré lemma (Theorem 3.4), and Stokes' theorem (Theorem 3.11). Deformation invariance (Corollary 3.13) is proved from Stokes, so it is proved modulo a cited theorem, and so is everything later that rests on it. Reparametrisation invariance (Proposition 3.9) is proved in a reading.
- *A deformation carries orientations.* Corollary 3.13 needs a deformation, not merely two loops that together bound a region. The circle $r = 2$ traversed anticlockwise and the circle $r = 1$ traversed clockwise bound the annulus between them as point sets, yet the angle form (3.4) integrates to 2π over the first and to -2π over the second.

Summary.

- *The exterior derivative.* $d \circ d = 0$ (Theorem 3.2), so exact forms are closed. The converse holds on a ball (the Poincaré lemma) and fails on the annulus (Proposition 3.6).
- *Integration.* A 1-form integrates over a curve with no further data and independently of the parametrisation, and $\oint df = 0$ round any closed curve.
- *Stokes* (cited) gives deformation invariance: a closed 1-form integrates to the same number over loops that deform into one another (Corollary 3.13). That is what will make the action variables well defined.
- *The symplectic form* is a closed, non-degenerate 2-form. On $\mathbb{R}^{2n}$ it is $\sum_i dq_i \wedge dp_i = -d(p \cdot dq)$, and its wedge order gives Hamilton's equations with the right signs (3.13).
- *Next:* Lecture 4 shows that the bracket and the forms are one structure, and finds the shape the invariant tori will have: Lagrangian.

Tutorial. Exercise Sheet 3.

References

da Silva, *Lectures on Symplectic Geometry*, §1 (symplectic forms) and §6 (forms, d , the Poincaré lemma, Stokes); Arnold, *Mathematical Methods of Classical Mechanics*, §§35–36 (integration of forms, Stokes), §37 (symplectic structures) and §44 (the action integral this lecture was built to support).